EN3160 Tutorial 1

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1. (a) Figure 1 shows an image and its gamma-corrected version.
   i. What could have been the value of gamma out of $0.5, 1.0, 1.5$?
   ii. Justify the answer.
   iii. Notice the windows in the picture. There was no visual enhancement when the same operation was used for an image with a bright large window. What are the possible reasons?

   (b) Fig. 2 shows a 6×8 image to be zoomed by a factor of 3.
   i. Compute the value of the resulting zoomed image at $(10, 14)$ using the nearest neighbor inerpolation.
   ii. Compute the value of the resulting zoomed image at $(10, 14)$ using the bilinear interpolation.
   iii. The resulting image produced by bilinear interpolation needs to be slightly smaller than the nearest neighbor method. What is the reason for this?

   (c) i. Generate a 1-D Gaussian kernel of length 3 with $\sigma = 1$.
   ii. Hence compute a 3×3 Gaussian kernel with $\sigma = 1$.
   iii. Guess the rank of the 3×3 kernel in 1(c)ii. Justify.
   iv. Multiple runs of the 3×3 kernel in 1(c)ii results in better smoothing. If this kernel is run twice, what is expected resultant σ?
   v. What is the reason for not getting the effect of this expected resultant σ?

2. (a) State an example each of
   i. Intensity transformations, and
   ii. Spatial filtering.



(a) Original image. (b) Gamma corrected image.

Figure 1: Gamma correction for 2b.

	0	1	2	3	4	5	6	7
0	0	36	146	36	25	36	0	0
1	36	146	219	182	73	0	36	36
2	0	36	109	73	120	0	36	73
3	0	36	109	109	222	146	182	182
4	0	0	75	36	109	182	219	73
5	0	0	10	22	73	73	73	36

Figure 2: Image for zooming 2b.

(b) Consider the intensity transformation given by

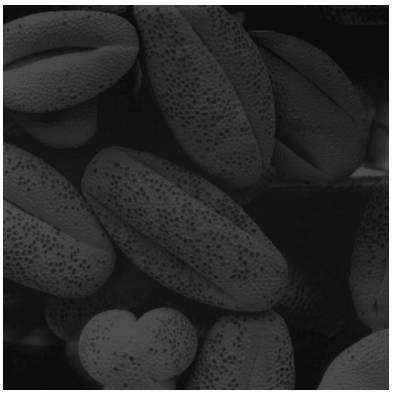
$$s = c\ln(r), \quad r \in [0, 255],$$

where r is the input intensity, s is the output intensity, and c is a constant to be estimated.

i. State the effect of this transformation.

ii. In a program, what is the usual way of applying a transform of this nature?

iii. A student's implementation of this transformation resulted in Fig. 3. What is the reason for the incorrectness?



(a) Original image.



(b) Intensity transformed image.

Figure 3: Intensity transformation 2b.

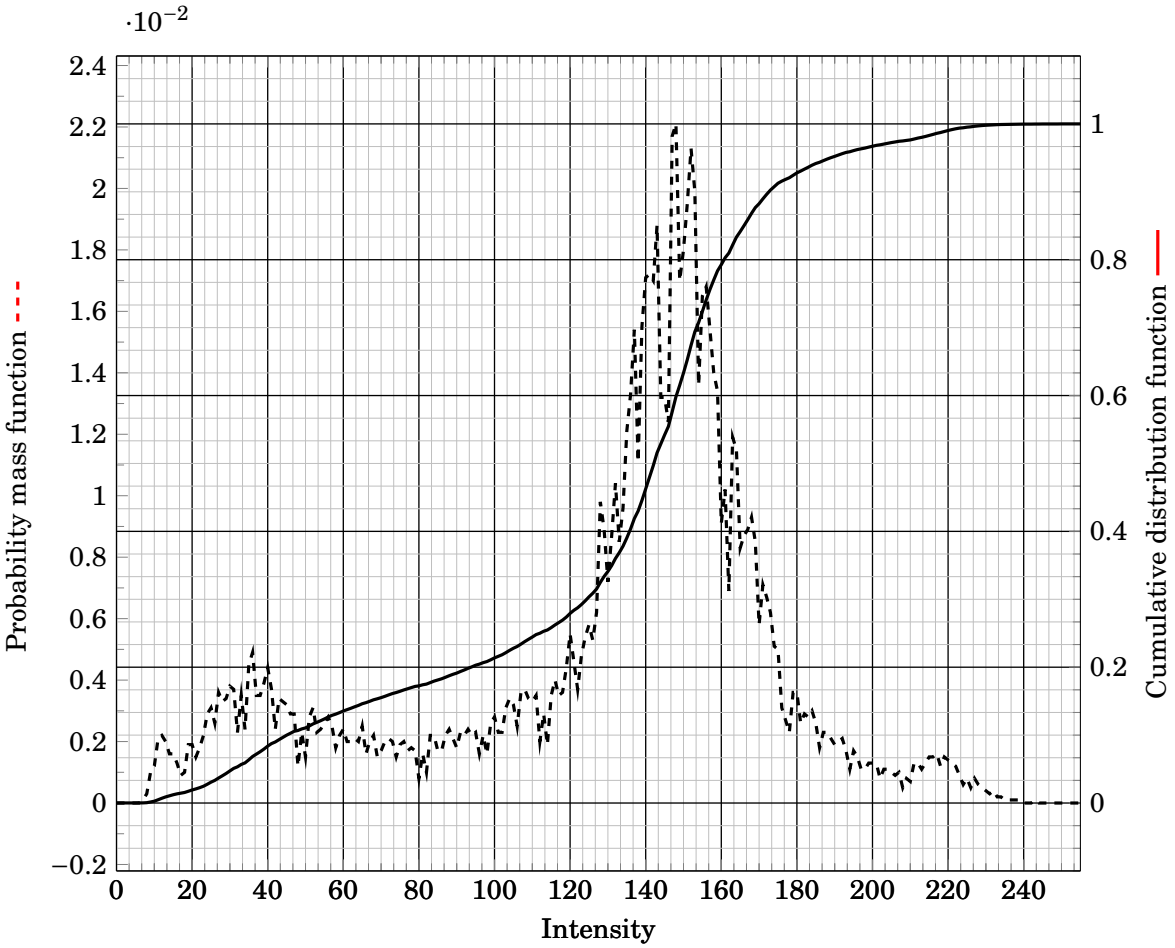


Figure 4: Intensity transformation 2c.

(c) Figure 4 shows the probability mass function $p_I(.)$ and cumulative distribution function of the intensities I of an 8-bit image. The darkest pixels (10%) and the brightest pixels (10%) are not important in this particular image. Consider the intensity transformation

$$s(i) = \begin{cases} 0, & \text{if } i \le L, \\ \dfrac{(i-L)255}{U-L}, & \text{if } L < i < U, \\ 255, & \text{if } i \ge U, \end{cases}$$

where

$$\sum_{0}^{L} p_I(i) = 0.1, \quad \text{and} \quad \sum_{U}^{255} p_I(i) = 0.1.$$

i. Briefly explain the effect of this transformation.

ii. Sate the values of L and U.

iii. Compute the resulting values of pixels with intensity $I_1 = 100$ and $I_2 = 200$?

(d) The Laplacian for a function $f(x,y)$ is

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}.$$

Discretization of this can result is a 3×3 Laplacian kernel.

i. Use the forward difference formula and backward difference formula together to obtain a symmetric expression for $\dfrac{\partial^2 f}{\partial x^2}$, if f is a digital image.

ii. Hence, write an expression for the Laplacian $\nabla^2 f$, if f is a digital image.

iii. Show this 3×3 Laplacian as a kernel.

iv. Figure 5 shows a different version of the 3×3 Laplacian kernel and an image. What is the resulting value at $(1,1)$ of the Laplacian image? Top-left pixel is $(0,0)$.

v. If the image is zero padded, what would be the sum of values of the resulting Laplacian image?

1	1	1
1	−8.0	1
1	1	1

(a) Kernel.

10	50	200	190	240
20	50	190	220	220
40	170	180	200	240
140	190	180	200	250

(b) Image.

Figure 5: Intensity transformation 2d.

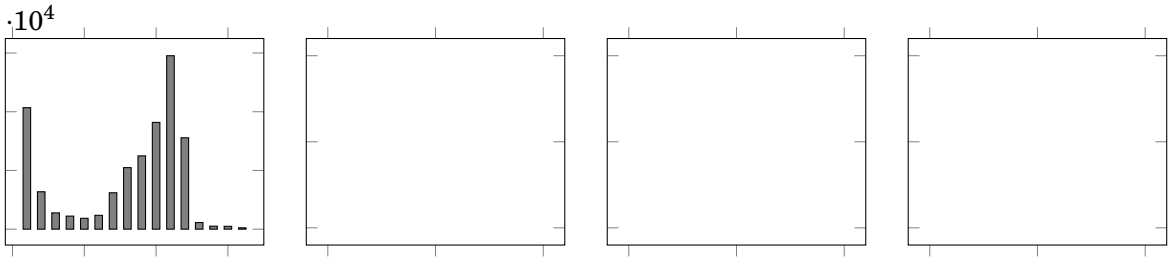
3. (a) Fig. 6 shows an image of a cameraman, its histogram, and three processed versions.

i. Approximately sketch the histograms of the processed images.

ii. State the probable image processing operation done on the source image to produce each of the three results shown in Fig. 6b, 6c, and 6d



(a) Original. (b) Processed 1. (c) Processed 2. (d) Processed 3.



(e) Histogram. (f) Answer. (g) Answer. (h) Answer.

Figure 6: Images and histograms.

(b) Figure 7 shows an intensity transfer function and an input image.

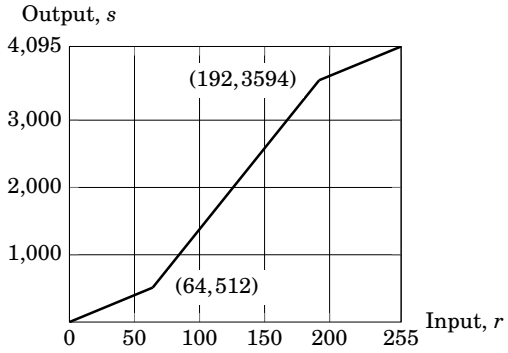
i. Judge the purpose of this transformation.

ii. Apply this transfer function to the input image and show the resulting pixel values in the grid shown.

iii. The *output* of the transformation in 3(b)ii is transformed using the operation represented by

$$s = c \ln(r), \quad r \in [0, 255],$$

where c is a constant. Discuss the effect of this.



(a) Transfer function.

150	200	255
100	100	200
50	120	100
10	10	50

(b) Input image.

(c) Grid for output.

Figure 7: Intensity transfer function, input image, and grid for output.

(c) The Laplacian of Gaussian is

$$l(x,y) = -\frac{1}{\pi\sigma^4}\left[1 - \frac{x^2+y^2}{2\sigma^2}\right] e^{-\frac{x^2+y^2}{2\sigma^2}}.$$

Figure 8 shows a part of the un-normalized 3×3 Laplacian of Gaussian kernel with $\sigma = 1.4$.

0.0243		

Figure 8: A part of the 3×3 Laplacian of Gaussian kernel.

i. Fill the other entries in the kernel.
ii. Why are all the entries positive?

(d) Fig. 9 shows an image, the foreground extraction, and the result with a blurred background.

i. Suggest an existing algorithm to extract the foreground.
ii. List the steps for producing the result given the original image and the foreground image (or foreground mask).
iii. Why is the background just beyond the edge of the flower dark in the enhanced (result) image?

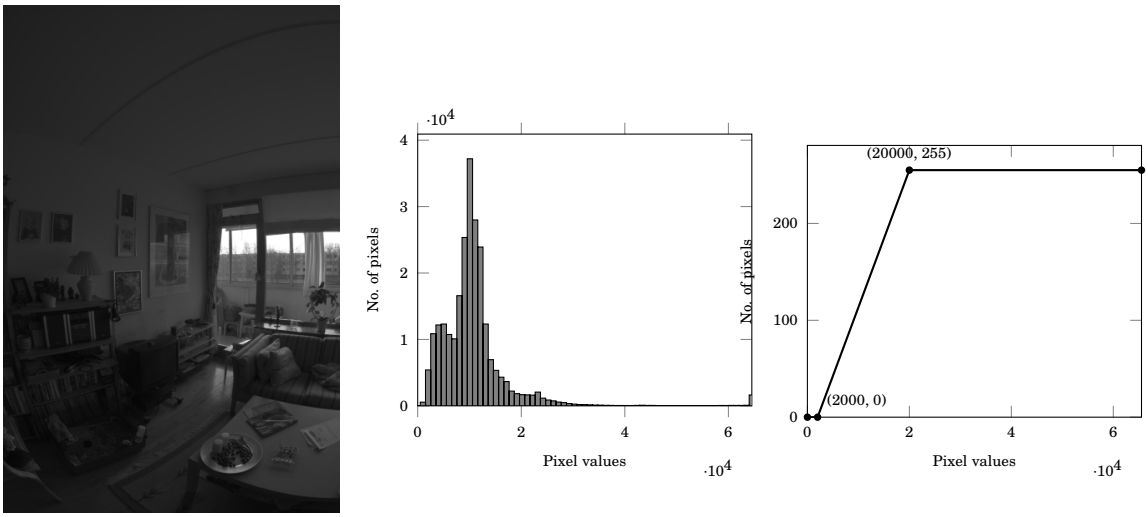
4. (a) Fig. 10 shows a 16-bit image (high dynamic range, HDR), its histogram, and a transformation representing a point operation. Note that this image cannot be displayed in a regular monitor and does not appear well in print in this paper.

i. Compute the value of pixels with intensity 10,000 and 50,000 due to this transformation.
ii. What would be the effect of applying this transformation?
iii. Sketch a transformation to make the details seen through the window clearer.



(a) Original image. (b) Foreground. (c) Result.

Figure 9: Original image, foreground image, result with blurred background.



(a) Original HRD image. (b) Histogram. (c) Transformation.

Figure 10: 16-bit image, its histogram, and a transformation.

(b) Derivative of Gaussian is a common filter used to detect edges.

i. Construct a 1-D Gaussian kernel with $\sigma = 1$ and size 7.

ii. An approximation to the above kernel is $\begin{bmatrix}0 & 1 & 5 & 8 & 5 & 1 & 0\end{bmatrix}/50$. Using this approximation, construct a 2-D 7×7 Gaussian kernel.

iii. Obtain the derivative of the 1-D kernel in 4(b)ii using the central difference approximation.

iv. Hence obtain the 2-D derivative of Gaussian kernel for the vertical direction.

(c) Fig. 11 shows an original image, an image with a shadow effect added to the foreground objects and intermediate images. Foreground objects are the maple leaves.

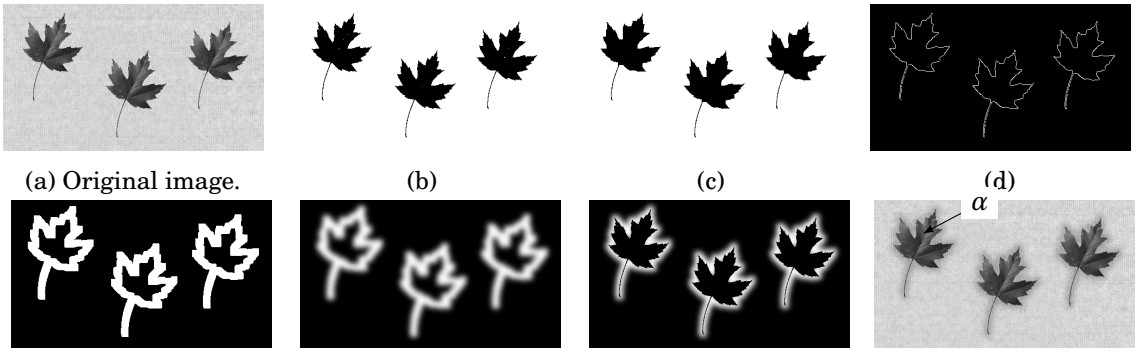
i. Identify each image in the sequence in Fig. 11 from the following list:

A. Otsu thresholded image,

B. Opened Otsu thresholded image,

C. Edges,

D. Edges run on the opened Otsu thresholded image,



(a) Original image. (b) (c) (d)

(e) (f) (g) (h) Shadow effect.

Figure 11: Adding a shadow-effect to foreground objects

E. Dilated edges,
F. Blurred edges,
G. Blurred dilated edges,
H. Blurred dilated edges masked with the Otsu thresholded image,

and fill the Table 1. Note that as the sequence of operations is what matters, the appearance of these images as printed does not detrimentally affect your answer.

Image	Operation (write the correct uppercase letter only)
11a	Original image
11b	
11c	
11d	
11e	
11f	
11g	
11h	Image with shadow effect added to foreground objects

Table 1: Sequence of operations.

ii. What is the reason for carrying out morphological opening on the Otsu thresholded image?

iii. The shadow near concave corners such as the one marked as α in Fig. 11h have a lighter shadow. Explain the reason for this.